Marking Schemes

This document was prepared for markers' reference. It should not be regarded as a set of model answers. Candidates and teachers who were not involved in the marking process are advised to interpret its content with care.

Chemistry
Paper 1

SECTION A

Question No.	Key	Question No.	Key
Part I		**Part II**	
1.	B (56%)	25.	D (76%)
2.	C (70%)	26.	B (71%)
3.	D (78%)	27.	A (57%)
4.	D (60%)	28.	B (69%)
5.	C (80%)	29.	B (48%)
6.	B (65%)	30.	D (83%)
7.	B (68%)	31.	B (43%)
8.	C (82%)	32.	D (45%)
9.	A (59%)	33.	A (58%)
10.	A (63%)	34.	C (55%)
11.	D (50%)	35.	A (59%)
12.	B (79%)	36.	C (65%)
13.	B (75%)		
14	C (49%)		
15.	C (83%)		
16.	A (65%)		
17.	D (42%)		
18.	A (66%)		
19.	D (68%)		
20.	A (63%)		
21.	B (41%)		
22.	A (72%)		
23.	C (47%)		
24.	C (48%)		

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

General Marking Instructions

1. In order to maintain a uniform standard in marking, markers should adhere to the marking scheme agreed at the markers' meeting.

2. The marking scheme may not exhaust all possible answers for each question. Markers should exercise their professional discretion and judgment in accepting alternative answers that are not in the marking scheme but are correct and well reasoned.

3. In questions asking for a specified number of reasons or examples etc. and a candidate gives more than the required number, the extra answers should not be marked. For instance, in a question asking candidates to provide two examples, and if a candidate gives three answers, only the first two should be marked.

4. In cases where a candidate answers more questions than required, the answers to all questions should be marked. However, the excess answer(s) receiving the lowest score(s) will be disregarded in the calculation of the final mark.

5. Award zero marks for answers which are contradictory.

6. Chemical equations should be balanced except those in reaction schemes for organic synthesis. For energetics, the chemical equations given should include the correct state symbols of the chemical species involved.

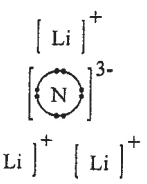
7. In the question paper, questions which assess candidates' communication skills are marked with an asterisk (*). For these questions, the mark for effective communication (1 mark per question) will be awarded if candidates can produce answers which are easily understandable. No marks for effective communication will be awarded if the answers produced by candidates contain a lot of irrelevant materials and/or wrong concepts in chemistry.

SECTION B

Part I

Marks

1. (a) (i) $6.0\,x + 7.0\,(1-x) = 6.9$ 2
$x = 0.1 = 10.0\%$

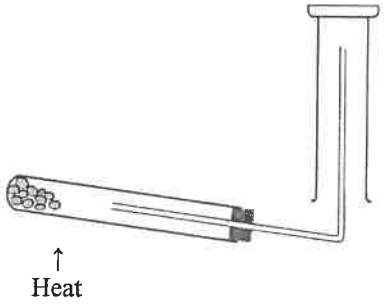
(ii)  1

(b) (i) $6Li + N_2 \rightarrow 2Li_3N$ 1

(ii) $y / 6.9 = 3 \times (1.25 / 34.7)$ 2
$y = 0.746$ g

(c) lithium oxide / lithium peroxide 1

2. (a) 2



(b) (i) Ammonia is soluble in water / Ammonia reacts with water to form aqueous ammonia. 1
As all ammonia dissolves, the atmospheric pressure forces the water in the trough to inject into the flask through the glass tube. 1

(ii) The water in the flask turns from colourless to pink. 1
It is because aqueous ammonia is alkaline. 1

Marks

3. (a) The electrostatic attraction between Ba^{2+} and Cl^- in $BaCl_2$ is ionic bond while intermolecular attraction between OCl_2 molecules are van der Waals' forces. / $BaCl_2$ is an ionic compound while OCl_2 has a simple molecular structure. 1

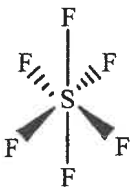
As ionic bond is much stronger than van der Waals' forces / intermolecular forces between OCl_2 molecules, $BaCl_2$ would have a higher melting point than OCl_2. 1

(b) • Both molecules of PH_3 and CH_4 are held by van der Waals' forces / intermolecular forces. 1

• The van der Waals' forces between PH_3 are stronger that that between CH_4 because of the larger molecular size of PH_3 than CH_4. / Intermolecular forces between PH_3 molecules are stronger than that between CH_4 molecules as PH_3 is polar while CH_4 is non-polar. 1

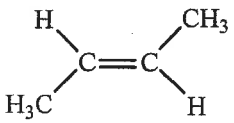
• Hydrogen bond exists among NH_3 molecules that is stronger than van der Waals' forces. 1

(c) 1



4. (a) Petroleum is formed when large quantities of dead marine organisms, such as planktons and algaes, that are buried underneath sedimentary rock and subjected to intense heat and pressure for a long time. 2

(b) (i) 1



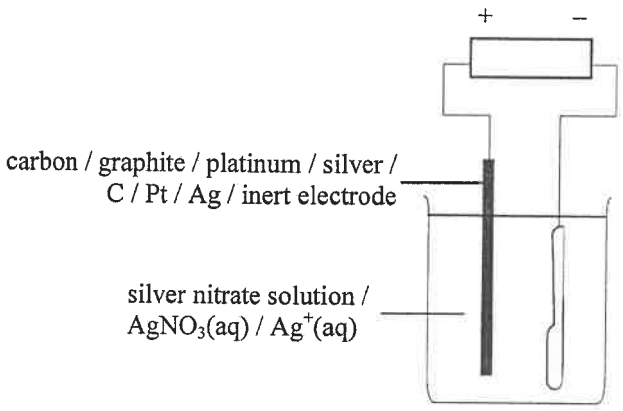
(ii) but-1-ene or methylpropene 1

(c) (i) Pass excess H_2 to ethene in the presence of Pt / Pd / Ni. / catalytic hydrogenation 1

(ii) Ethene turns Br_2(in CH_3CCl_3) from brown / orange to colourless while ethane does not. 2

Marks

5. (a)



2

(b) • Connect zinc/magnesium blocks to the surface of the pipelines. / sacrificial protection 1
• Zinc/Magnesium can release electrons more readily than iron. 1
OR
• Connect the negative electrode of a D.C. source to the surface of the pipelines. / cathodic protection
• The electrons provided by the D.C. source prevent iron from releasing electrons.

6. (a) (i) $6CO_2(g) + 6H_2O(l) \rightarrow C_6H_{12}O_6(s) + 6O_2(g)$ 1

(ii) $\Delta H = -1274 - 6 \times (-394-286)$ 2
$= +2806$ kJ mol^{-1}

(iii) Light energy changes to chemical energy. 1

(b) (i) Let C be the heat capacity of the calorimeter. 3
$-715 \times (1.58 / 32.0) = -C \times 18.5$
$\Delta H \times (1.02 / 100.0) = -C \times 25.8$
$\Delta H = -4826.8$ kJ mol^{-1}

(ii) Some methanol or heptane evaporates. / incomplete combustion 1

7. (a) conical flask 1

(b) yellow to orange 1

(c) Number of moles of $B_4O_7^{2-}(aq) = 0.125 \times 0.01898 \times ½ = 1.187 \times 10^{-3}$ 3
$(201.2 + 18n) = 0.452 / 1.187 \times 10^{-3}$
$n = 10$

(d) (i) solutions with accurately known concentrations 1

(ii) It can be used to determine the concentration of another reagent / number of water of crystallisation / molar mass, etc. 1

Marks

8. (a) An acid which can almost completely ionise in water. 1

(b) (i) chlorine / $Cl_2(g)$ 1

(ii) It is a redox reaction: oxidation number of Cl changes from -1 to 0 / of Mn changes from +7 to +2 / Cl^- transfers electrons to MnO_4^- / MnO_4^- is reduced and Cl^- is oxidised. 1

(c) The filter paper turns brown. 1
$2I^- + Cl_2 \rightarrow 2Cl^- + I_2$ 1

(d) The experiment should be performed in a fume cupboard as chlorine gas is toxic. 1

9. Chemical knowledge (1 mark for each point, a maximum of 4 marks) 4

- Compounds with C=C bonds can undergo addition polymerisation.
- No small molecules will be eliminated during addition polymerisation.
- High temperature / High pressure / Catalyst is used.
- Structure of the monomer : $CF_2=CF_2$
- Structure of the repeating unit : $-CF_2-CF_2-$ OR the polymer : $-[CF_2-CF_2]_n-$

Communication mark 1

Part II

Marks

10. Step One: (1) $LiAlH_4$ (2) H_3O^+ 1
Intermediate: $HOCH_2CH_2CH_2CH_2CH_2OH$ 1
Step Two: PCl_3 / PCl_5 / HCl / $SOCl_2$ 1

11. (a) (i) colour intensity / absorbance 1

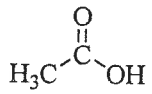
(ii) On the graph, plot a tangent at time = 0 on the curve. 1
The initial rate equals to the slope of the tangent. 1

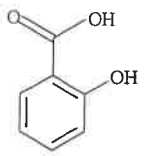
(iii) The absorbance is directly proportional to $[Br_2(aq)]$ / number of Br_2 molecules in the reaction mixture. The $[Br_2(aq)]$ / number of Br_2 molecules in the reaction mixture at **A** is higher than that at **B**. Therefore the frequency of effective collisions between molecules at **A** is higher than that at **B**. 2

(b) At different time, measure the volume of CO_2 gas formed / total pressure of the system / mass of the reaction mixture. 1

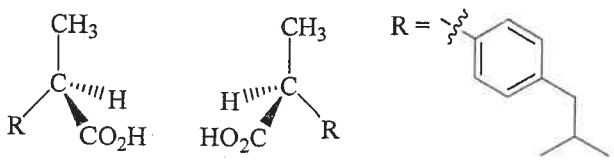
12. (a) Reduce fever / inflammation / risk of heart disease. 1

(b) -COOH group of aspirin reacts with hydrogencarbonate ions in water to give a soluble sodium salt / soluble ions / soluble $-COO^-$. 2

(c) (i)  1

 1

(ii) Hydrolysis of ester in acidic medium is reversible. Equilibrium will be attained if the reaction mixture is heated under reflux for a long time. 1

(d)  2

Marks

13. (a) None of the final concentrations of **X**(g), **Y**(g) and **Z**(g) is equal to zero. / 1
X(g), **Y**(g), **Z**(g) co-exist in the system, and their concentrations remain unchanged after a long period of time. /
The concentration of the reactant **Y**(g) is still not equal to zero after a long period of time.

(b) $2\mathbf{Y}(g) \rightleftharpoons 3\mathbf{X}(g) + \mathbf{Z}(g)$ 3

$Kc = [\mathbf{X}(g)]^3 [\mathbf{Z}(g)] / [\mathbf{Y}(g)]^2$
$= (0.60)^3 (0.20) / (0.30)^2$
$= 0.48 \text{ mol}^2 \text{ dm}^{-6}$

(c) The statement is incorrect. At the 25th minute after the reaction has started, the reaction attained dynamic equilibrium. The rate of forward reaction is equal to the rate of backward reaction and not equal to zero. 1

14. Chemical knowledge 5

- $Na_2O(s)$ dissolves in water to give NaOH(aq) /
  $Na_2O(s)$ reacts with HCl(aq) to give NaCl(aq) and $H_2O(l)$ /
  $Na_2O(s) + H_2O(l) \rightarrow 2NaOH(aq)$ /
  $Na_2O(s) + 2HCl(l) \rightarrow 2NaCl(aq) + H_2O(l)$
- $Al_2O_3(s)$ reacts with HCl(aq) to give $AlCl_3(aq)$ and $H_2O(l)$ /
  $Al_2O_3(s) + 6HCl(aq) \rightarrow 2AlCl_3(aq) + 3H_2O(l)$
- $Al_2O_3(s)$ reacts with NaOH(aq) to give $NaAl(OH)_4(aq)$ /
  $Al_2O_3(s) + 2NaOH(aq) + 3H_2O(l) \rightarrow 2NaAl(OH)_4(aq)$ /
  $Al_2O_3(s) + 2NaOH(aq) \rightarrow 2Na[AlO_2](aq) + H_2O(l)$
- $SO_2(g)$ dissolves in water to give $H_2SO_3(aq)$ /
  $SO_2(g)$ reacts with NaOH(aq) to give $Na_2SO_3(aq)$ and $H_2O(l)$ /
  $SO_2(g) + H_2O(l) \rightarrow H_2SO_3(aq)$ /
  $SO_2(g) + 2NaOH(aq) \rightarrow Na_2SO_3(aq) + H_2O(l)$
- Na_2O is a basic (alkaline) oxide, Al_2O_3 is an amphoteric oxide, and SO_2 is an acidic oxide.

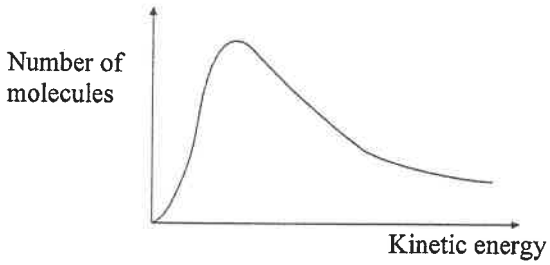
Communication mark 1

Paper 2

Marks

1. (a) (i) $2H^+(aq) + 2e^- \rightarrow H_2(g)$ 1
OR $2H_2O(l) + 2e^- \rightarrow 2OH^-(aq) + H_2(g)$
$2Cl^-(aq) \rightarrow Cl_2(g) + 2e^-$ 1

(ii) 2



(iii) glucose 1

(b) (i) (1) Cu / ZnO / Al_2O_3 1

(2) May be due to higher activation energy. 1

(3) Number of moles of gaseous products is less than that of gaseous reactants. 1
Increasing the pressure will shift the equilibrium position to the right. 1

(ii) (1) This reaction does not involve poisonous reagent but the original one involves poisonous CO. 1

(2) It reduces the amount of atmospheric carbon dioxide and hence may alleviate global warming. 1

(iii) $CH_3OH(g) + CO(g) \rightarrow CH_3CO_2H(g)$ 1

(c) (i) Order of reaction is not affected by temperature change. 1

(ii) From line ℓ_1 on the graph, 2
slope = [(−1.4) − (−2)] ÷ [(0)−(−0.6)] = 1
It is first order with respect to $N_2O_5(g)$.

(iii) $log\ k = -2$ 2
$k = 0.01\ s^{-1}$

(iv) The y-intercepts of ℓ_1 and ℓ_2 are −1.4 and −2 respectively. 3
Since the y-intercept = $log\ k$,
(−2) − (−1.4) = Ea (1/360 − 1/345) / 2.3 x 8.31
Ea = 94.95 kJ mol^{-1}

Marks

2. (a) (i) • Catalyst is used. 1
• High atom economy / atom economy = 172 ÷ 208 = 82.7% 1

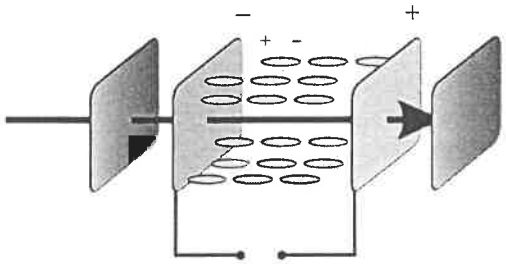
(ii) (1) chitin 1

(2) It can form extensive inter-molecular hydrogen bonds. 1

(iii) **A** 1

(b) (i) cholesteric liquid crystal 1
The molecules are arranged along a long axis and in positions such that they are slightly twisted from the molecules next to them giving rise to a helical-like arrangement. 1

(ii) (1)



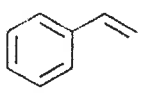
voltage applied

Show: With the applied voltage, the molecules arrange in lines. 1
Show: The polarity of the molecules and that of electrodes are opposite. 1

(2) Polarisers are perpendicular to each other. 1
The polarised light passes through the liquid crystal layer without rotating the plane of polarisation. The polarised light is completely blocked by the polariser at the right, giving a dark pixel. 1

(iii) When temperature is higher than the upper end of the operation range, the liquid crystal will liquefy. 1

(c) (i) $HOCH_2CH_2OCH_2CH_2OH$ 1
$HOOC–CH=CH–COOH$ / $Cl–CO–CH=CH–CO–Cl$ 1

(ii) (1)  1

(2) thermosetting / hardening under formation with heating 1

(3) Adjust the relative amounts of **X** and **A**. 1
The rigidity depends on its degree of cross-linking. 1

(iii) (1) compression moulding 1

(2) **Y** will not corrode easily but iron will. / **Y** is less dense than iron. 1

Marks

3. (a) (i) calcium ion / Ca^{2+} 1

(ii) Add $Ba(NO_3)_2(aq)$ acidified with $HNO_3(aq)$ to the sample, 1
white precipitate forms for $K_2SO_4(aq)$ whereas no precipitate will form for $K_2SO_3(aq)$. 1

(iii) 'R_f value' of a substance is the ratio between the migration distance of the substance and the migration distance of the solvent front during chromatography. 2

(b) (i) Place the dissolved sample into a 250.0 cm^3 volumetric flask. 1
Deionised water should be added to the mark of the volumetric flask. Shake the solution thoroughly. 1

(ii) $ClO_3^-(aq) + 6I^-(aq) + 6H^+(aq) \rightarrow Cl^-(aq) + 3I_2(aq) + 3H_2O(l)$ 1

(iii) The solution turns from blue to colourless. 1

(iv) $I_2(aq) + 2S_2O_3^{2-}(aq) \rightarrow 2I^-(aq) + S_4O_6^{2-}(aq)$ 3
number of moles of $I_2(aq) = 0.112 \times 0.02788 \times ½ = 0.001561$
number of moles of $NaClO_3$ in the sample $= 0.001561 \div 3 \times (250.0 / 10.00) = 0.01301$
percentage by mass of $NaClO_3$ in the sample
$= 0.01301 \times 106.5 \div 1.63 \times 100\% = 85.0\ \%$

(c) (i) Boiling points of **X** and **Y** are too close. 1

(ii) (1) Absorption peak at wavenumber about 1700 cm^{-1} corresponds to a C=O group. / 1
Absorption peak at wavenumber about 1650 cm^{-1} corresponds to a C=C group.

(2) At m/z = 43: CH_3CO^+ 1
At m/z = 55: CH_2CHCO^+ 1

(3) $CH_2{=}CHCOCH_3$ 1

(iii) • positive result for 2,4-dinitrophenylhydrazine test: presence of carbonyl group 1
• negative result for Tollens' reagent test: not an aldehyde 1
• **Y** may be $CH_3CH_2COCH_3$ / butanone. 1

Candidates' Performance

Paper 1

Paper 1 consisted of two sections, Section A (multiple-choice questions) and Section B (conventional questions). Sections A and B each comprised two parts, Part I and Part II. Part I contained questions mainly on Topics I to VIII of the curriculum, while Part II mainly on Topics IX to XII. All questions in both sections were compulsory.

Section A (multiple-choice questions)

This section consists of 36 multiple-choice questions. The mean score was 22.5. Candidates' performance was generally good. Some misconceptions of candidates were revealed from their performance in the following items:

1. For Q.11, it should be noted that the $CH_3CO_2H(aq)$ in solution **Y** is a weak acid that does not completely ionise into hydrogen ions. The $[H^+(aq)]$ should be less than 0.20 M, and hence the pH of **Y** does not equal –log 0.2. Option A is therefore not correct. For option C, as both solutions **W** and **X** contain strong acids of equal basicity and concentration, the pH of them would be the same. Mixing of them would not give a change in pH of the mixture. That is why this option is also wrong.

Q.11 Consider the solutions **W**, **X**, **Y** and **Z** below :

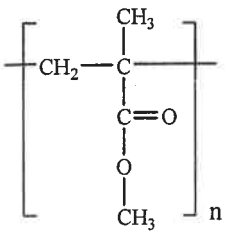
W	100 cm^3 of 0.20 M $HNO_3(aq)$
X	50 cm^3 of 0.20 M $HCl(aq)$
Y	100 cm^3 of 0.20 M $CH_3CO_2H(aq)$
Z	50 cm^3 of 0.10 M $NaOH(aq)$

Which of the following statements is correct ?

A.	The pH of **Y** equals –log 0.2.	(20%)
B.	Mixing **W** and **Z** gives a neutral solution.	(11%)
C.	The pH of the mixture of **W** and **X** is lower than that of **W**.	(19%)
D.*	The pH of the mixture of **W** and **X** is lower than that of the mixture of **X** and **Y**.	(50%)

2. For Q.14, 32% of candidates wrongly chose option A as the answer. As a polyester should have the ester groups of the repeating units appear in the mother chain of the polymers and from the structure given in the item, the ester groups only appear in the branch chain of the repeating units. This explains why the polymer is not a polyester.

Q.14 A polymer has the following structure :



Which of the following statements concerning the polymer is correct ?

A.	It is a polyester.	(32%)
B.	It can be polymerised from $(CH_3)_2CHCO_2CH_3$.	(8%)
C. *	Its monomer can decolourise acidified $KMnO_4(aq)$.	(49%)
D.	It can be made from its monomer through condensation.	(11%)

3. For Q.32, other than those correctly choosing the key D as answer, many candidates wrongly chose B instead of D showing that they mistakenly considered (3) cannot illustrate the characteristics of transition metals. In fact $CuSO_4(s)$ is white and turns blue when mixed with $H_2O(l)$, that can serve as an example of 'most aqueous ions of transition metals are coloured'.

Q.32 Which of the following processes can illustrate the characteristics of transition metals ?

(1) mixing $AgNO_3(aq)$ and $NaCl(aq)$
(2) mixing $FeSO_4(aq)$ and $Br_2(aq)$
(3) mixing $CuSO_4(s)$ and $H_2O(l)$

A.	(1) only	(12%)
B.	(2) only	(34%)
C.	(1) and (3) only	(9%)
D.*	(2) and (3) only	(45%)

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B.	(2) only	(34%)
C.	(1) and (3) only	(9%)
D.*	(2) and (3) only	(45%)

Section B (conventional questions)

Question Number	Performance in General
1	The performance of candidates in this question was good. In part (a)(i), about three quarters of the candidates were able to calculate the correct answer for the percentage abundance of ^{6}Li in nature. In part (a)(ii), about half of the candidates were able to give the correct electron diagram for lithium nitride. A small number of candidates wrongly considered lithium nitride is a covalent compound and gave an incorrect electron diagram. In part (b), about three quarters of the candidates were able to write the correct chemical equation for the formation of lithium nitride, and calculate the mass of lithium that reacted with nitrogen correctly. In part (c), a high proportion of the candidates were able to give the correct name of the compound formed. A small number of candidates wrongly gave the chemical formula instead of the name as required by the question.
2	The performance of candidates in this question was satisfactory. In part (a), about a third of the candidates were able to give a correct labelled diagram for the experimental set-up. A small number of candidates wrongly drew a set-up that the ammonia gas is collected by displacement of water. In part (b)(i), less than half of the candidates were able to give the correct answer. Some candidates were able to state that ammonia is soluble in water, but failed to explain why the water is sucked into the flask. In part (b)(ii), just less than half of the candidates were able to state the correct colour change of phenolphthalein and give a correct explanation. A small number of candidates only stated the colour change in the answer without giving any explanation, or wrongly stated that the NH_4^+ ion is alkaline.
3	The performance of candidates in this question was satisfactory. In part (a), about half of the candidates were able to correctly explain why $BaCl_2$ has a higher melting point than OCl_2. Some candidates wrongly stated that ionic bonds exist between $BaCl_2$ 'molecules', or exist between Ba 'atoms' and Cl 'atoms'. In part (b), just less than half of the candidates were able to give a correct and complete answer. Quite a large number of candidates failed to state explicitly that hydrogen bond is stronger than van der Waals' forces. Some candidates wrongly stated that the van der Waals' forces between PH_3 are stronger than that between CH_4 because the molecular mass of PH_3 is higher than that of CH_4. In part (c), about two thirds of the candidates were able to give the correct three-dimensional diagram for the SF_6 molecule.
4	The performance of candidates in this question was good. In part (a), just less than half of the candidates were able to give the correct origin of petroleum in a complete way. Some candidates failed to mention that petroleum was formed from marine organisms. In part (b), a high percentage of the candidates were able to give the correct structure of the *trans*-isomer, and about two thirds of the candidates were able to give the correct systematic name of one possible structure of **F**. Some candidates wrongly gave methylprop-1-ene as the answer. In part (c)(i), about two thirds of the candidates were able to suggest a correct method to convert ethene into ethane. In part (c)(ii), about two thirds of the candidates were able to suggest a correct chemical test and state the expected observations correctly. Some candidates wrongly stated that ethane and ethene can be distinguished by using acidified potassium dichromate.
5	The performance of candidates in this question was good. In part (a), about two thirds of the candidates were able to complete the set-up using appropriate drawings and labels. In part (b), about one third of the candidates were able to describe a suitable method for protecting the pipelines, but only about a quarter of them correctly explained the principle behind the method described.

Question Number	Performance in General
6	The performance of candidates in this question was fair. In part (a), a very high proportion of the candidates were able to write the chemical equation involved. About half of the candidates correctly calculate the standard enthalpy change required. However, only about one third of the candidates were able to state the transformation of energy. Common mistakes included 'heat energy to chemical energy' and 'potential energy to chemical energy'. In part (b), just less than a quarter of the candidates were able to determine the enthalpy change required. It was encouraging that some candidates made use of proportion for completing the calculation and presenting the steps and answer in a concise manner.
7	The performance of candidates in this question was satisfactory. In part (a), about half the candidates were able to state **X** as a conical flask. Quite a number of them wrote incorrect answers like 'volumetric flask', 'beaker' or 'burette'. In part (b), about half the candidates were able to state the colour change at the end point of the titration. In part (c), about two thirds of the candidates were able to calculate the number of water of crystallisation. In part (d), about half of the candidates were able to describe the exact meaning of the term 'standard solutions'. However, quite a number of them incorrectly included 'known volume' in their answers.
8	The performance of candidates in this question was satisfactory. In part (a), about half of the candidates were able to describe the exact meaning of the term 'strong acid', while others gave incorrect answers like 'a solution with low pH', 'H^+(aq) completely ionises in water' or 'decomposed completely to give H^+(aq)'. In part (b), a very high proportion of the candidates were able to recognise the gas as 'chlorine'. In part (c), about half of the candidates were able to describe the expected observation but only about a third of them correctly wrote the ionic equation to represent the change. In part (d), only a third of the candidates were able to state and give the correct reason why the experiment should be carried out in a fume cupboard. Some of them wrongly considered chlorine as only harmful instead of poisonous.
9	The performance of candidates in this question was fair. About half of the candidates were able to only describe two pieces of information about the polymerisation. For example, they only stated that the polymerisation involved a carbon carbon double bond and no elimination of small molecules, but did not describe the structures of the reactant (monomer) and the product (polymer). A few candidates erroneously considered tetrafluoroethene as the repeating unit of the polymer. Some candidates were able to outline the reaction conditions required. In general, candidates were weak in presenting their answers in a systematic way.
10	The performance of candidates in this question was satisfactory. About a third of the candidates were able to give a complete and correct synthetic route, while many candidates made some mistakes in one or two of the synthetic steps involved. Some candidates failed to state the correct reaction conditions for using $LiAlH_4$ to reduce the starting material into the intermediate pentane-1,5-diol, or wrongly stated that the intermediate pentane-1,5-diol can be converted into the product by using Cl_2.

Question Number	Performance in General
11	The performance of candidates in this question was satisfactory. In part (a)(i), just below half of the candidates were able to state that the parameter is colour intensity or absorbance. Some candidates wrongly gave concentration of bromine as the answer. In part (a)(ii), a high proportion of the candidates were able to correctly state the initial rate of the reaction can be obtained from the slope of the tangent. However, some candidates failed to draw the tangent on the graph correctly. In part (a)(iii), about half of the candidates were able to explain the difference in rate between **A** and **B** in terms of effective collision frequency, but some of them failed to relate the parameter in the graph to the concentration of Br_2 in their answers. In part (b), about two thirds of the candidates were able to suggest an appropriate alternative method to follow the progress of the reaction.
12	The performance of candidates in this question was satisfactory. In part (a), about half of the candidates were able to state one correct medical application of aspirin other than as a pain-killer. In part (b), about a third of the candidates were able to give a correct and complete answer. Some candidates were able to suggest that the solution turns clear due to the reaction between aspirin and sodium hydrogencarbonate. However, they failed to state clearly that it is due to the reaction between the carboxyl group in aspirin and hydrogencarbonate which gives a soluble sodium salt. In part (c)(i), about half of the candidates were able to give the correct structures of the two organic products. In part (c)(ii), about half of the candidates were able to correctly state that the hydrolysis of aspirin is reversible. Some candidates wrongly stated that only dilute acid is used in the reaction, and the acid is used up before the reaction goes to completion. In part (d), about half of the candidates were able to give the correct three-dimensional structures for the pair of enantiomers. Some candidates made careless mistakes in drawing the substituents that connected to the chiral center, and some candidates only drew one enantiomer in the answer, or gave two identical structures.
13	The performance of candidates in this question was satisfactory. In part (a), just less than half of the candidates were able to explain that the reaction is reversible using the information provided in the graph. Some candidates merely stated that the reaction attained equilibrium without clear elaboration. In part (b), about forty percent of the candidates were able to calculate the equilibrium constant of the reaction correctly and answer with a correct unit. Some candidates failed to extrapolate the correct final concentrations of **X**(g), **Y**(g) and **Z**(g) from the graph, and some candidates failed to work out the correct stoichiometric relationships between **X**(g), **Y**(g) and **Z**(g) from the given data. In part (c), about two thirds of the candidates were able to give the correct answer.
14	The performance of candidates in this question was fair. About two thirds of the candidates were able to state that Na_2O is basic, Al_2O_3 is amphoteric while SO_2 is acidic. Some candidates were not able to give the correct reactions when illustrating the acid-base behaviour of the oxides. Common mistakes made include incorrect chemical formulae given for the reaction products such as $Na(OH)_2$; Na_2O reacts with water forming NaOH and H_2; Al_2O_3 dissolves in water forming alkaline $Al(OH)_3$, SO_2 dissolves in water forming H_2SO_4, and SO_2 reacts with NaOH forming Na_2SO_4 and water, etc. Many candidates failed to give the correct spelling of 'amphoteric', and some candidates wrongly stated that Al_2O_3 is an 'atmospheric' oxide.

Paper 2

Paper 2 consisted of three sections. Section A contained questions set on Topic XIII 'Industrial Chemistry', Section B on Topic XIV 'Materials Chemistry' and Section C on Topic XV 'Analytical Chemistry'. Candidates were required to attempt all questions in two of the sections.

Question Number	Popularity (%)	Performance in General
Section A: 1	98	The performance of candidates in part (a) was satisfactory. In part (a)(i), about two thirds of the candidates were able to write the two half equations for the electrolysis of brine using membrane electrolytic cell. In part (a)(ii), about half of the candidates were able to sketch a labelled diagram for a Maxwell-Boltzmann distribution curve. In part (a)(iii), just under half of the candidates were able to select glucose as the raw material for manufacturing vitamin C. The performance of candidates in part (b) was fair. In part (b)(i)(1), less than half of the candidates were able to suggest a suitable catalyst for the given reaction. In part (b)(i)(2), about half of the candidates were able to suggest a higher activation energy in the absence of a catalyst that would allow the reaction to proceed slowly. In part (b)(i)(3), about two thirds of the candidates were able to explain why the operation pressure in industry for the reaction is set at 100 atm but not at atmospheric pressure in terms of principles of equilibrium and kinetics. In part (b)(ii)(1), about a third of the candidates were able to comparatively show how Reaction (II) is greener than Reaction (I). In part (b)(ii)(2), about half of the candidates were able to suggest there is a potential of alleviating global warming for Reaction (II). In part (b)(iii), about a quarter of the candidates were able to write a chemical equation for the industrial production of ethanoic acid from methanol. The performance of candidates in part (c) was fair. In part (c)(i), about a third of the candidates were able to deduce that the order of the reaction is not affected by temperature change. In part (c)(ii), about two-thirds of the candidates were able to deduce the order of reaction with respect to N_2O_5. In part (c)(iii), less than half of the candidates were able to deduce the rate constant for the reaction with the appropriate unit. In part (c)(iv), about a third of candidates were able to calculate the activation energy of the reaction with the correct unit and appropriate number of significant figures.

Question Number	Popularity (%)	Performance in General
Section B: 2	6	The performance of candidates in part (a) was fair. In part (a)(i), only about a third of the candidates were able to explain using the appropriate green chemistry principles. In part (a)(iii), about two thirds of the candidates selected the correct answer. The performance of candidates in part (b) was very poor. Only a very small number of the candidates were able to state the name of the liquid crystal phase, draw a diagram to show the arrangement of the liquid crystal molecules and explain why the pixel appears dark when a voltage is applied. The performance of candidates in part (c) was poor. In part (c)(i), about a quarter of the candidates were able to draw the structures of the monomers of **X**. In part (c)(ii)(2), only a quarter of the candidates were able to provide a correct comment on the thermal property of **Y**. In part (c)(ii)(3), only a very small number of the candidates were able to put forward a correct way to control the rigidity of **Y** (i.e. control the relative amounts of **X** and **A**).
Section C: 3	96	The performance of candidates in part (a) was satisfactory. In part (a)(i), nearly all candidates were able to state the presence of calcium ion in the compound. In part (a)(ii) and (iii), only half of the candidates were able to suggest a chemical test to distinguish between $K_2SO_3(aq)$ and $K_2SO_4(aq)$ and suggest the meaning of R_f value in a paper chromatogram. The performance of candidates in part (b) was fair. In part (b)(i), about half of the candidates were able to describe how the sample dissolved in making up a 250.0 cm^3 solution. In part (b)(ii), (iii) and (iv), about a third of the candidates were able to write the ionic equation for the reaction involved, the colour change at the end point and calculate the percentage by mass of $NaClO_3(s)$ in the sample. The performance of candidates in part (c) was satisfactory. In part (c)(i), a very high proportion of the candidates were able to suggest the small difference in boiling points between **X** and **Y** making fractional distillation not a suitable method to separate **X** from the mixture. In part (c)(ii)(1), about half of the candidates were able to deduce the functional group that may be present in **X**. In part (c)(ii)(2) and (3), about a third of the candidates were able to suggest one chemical species for the corresponding peaks in the mass spectrum and draw CH_2=$CHCOCH_3$ as a possible structure of **X**. In part (c)(iii), about half of the candidates were able to deduce **Y** as butanone.